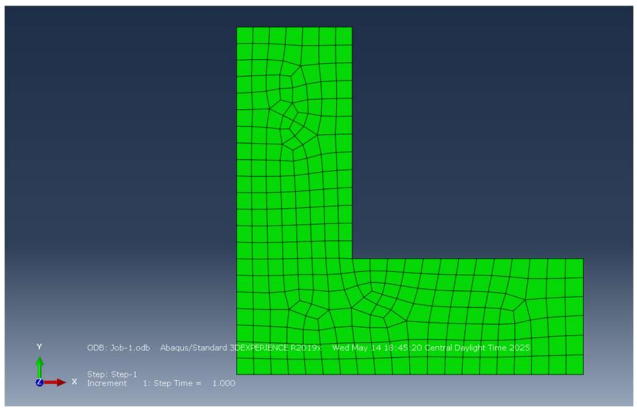
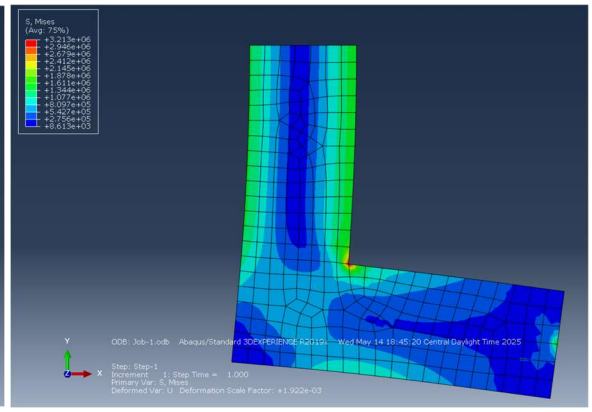
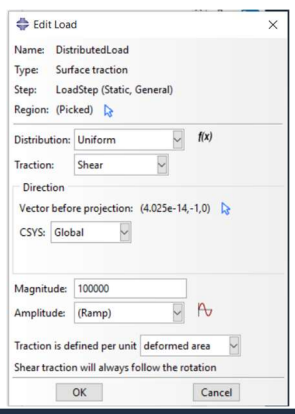
**Problem :**

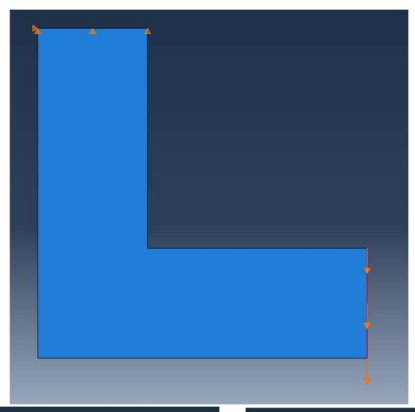
For this problem, I chose to use a reduced integration Q8 element, for better stress/strain accuracy than a triangle element. The top of the element and the top left corner were constrained as instructed, and a point load was administered on the right side.

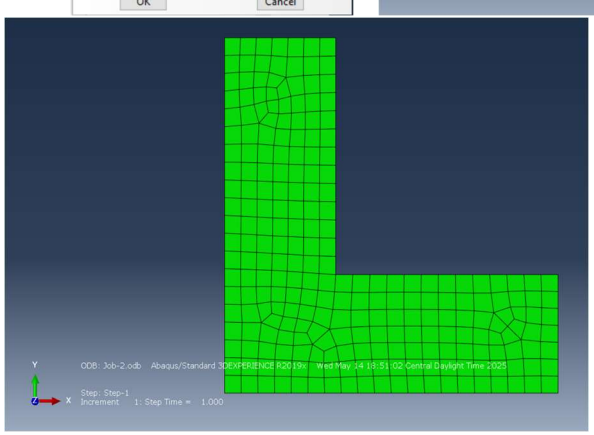


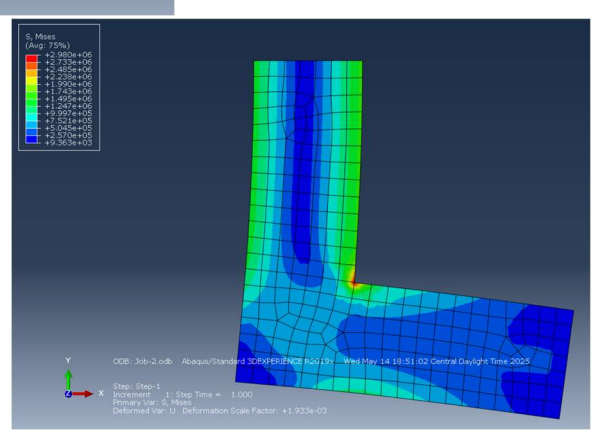




The maximum Von Mises stresses don't surpass the yield stress, so the given load is permissible. The region to monitor carefully is the inside bottom left corner, due to the high stresses it endures. We now apply a statically equivalent load, but make it evenly distributed along the right side, instead of point load. The new load was calculated by dividing the original load by the length of the side, and implemented in ABAQUS as follows, with the mesh and result below.



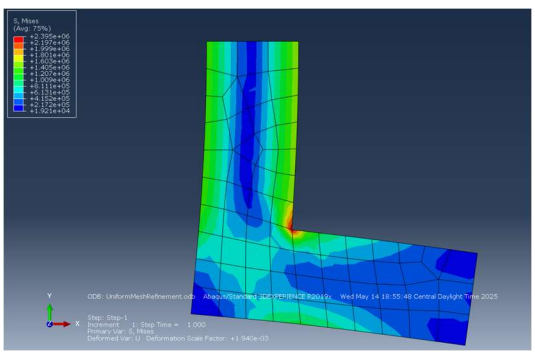


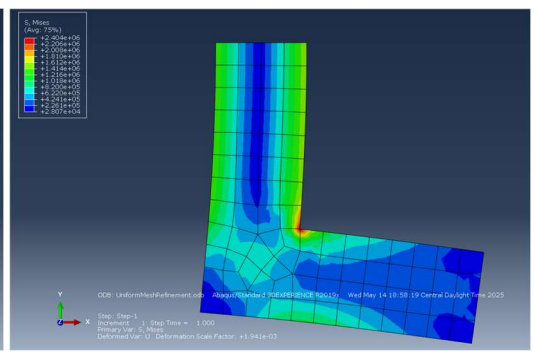



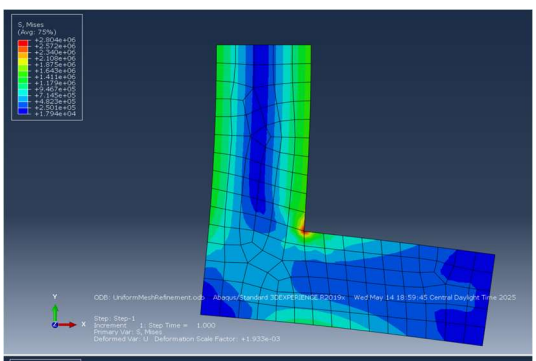
When the point load is replaced by a distributed load of equal (total) magnitude, the localized stress singularity near the load application site disappears, resulting in a smoother and more realistic stress distribution. According to Saint-Venant's Principle, differences in stress caused by different, but statically equivalent loads, become negligible at distances sufficiently far from the loaded region. This is confirmed in our simulation; far from the right edge, the stresses and strains under both loading conditions are nearly identical, validating the principle's prediction that local load details don't affect the global response significantly.

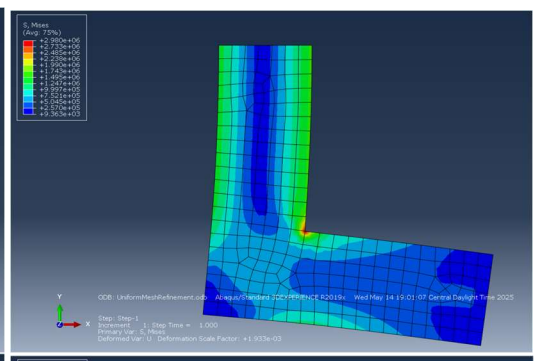
Fixing the top left corner node in both directions provides a stable and unique reference point for the numerical solution.  If we don't do this, the body could experience rigid body motion, meaning it could slide or rotate slightly in the solution without violating constraints. By anchoring the top-left corner in both directions, we eliminate this possibility and ensure a well-posed, solvable system with no rigid body motion.

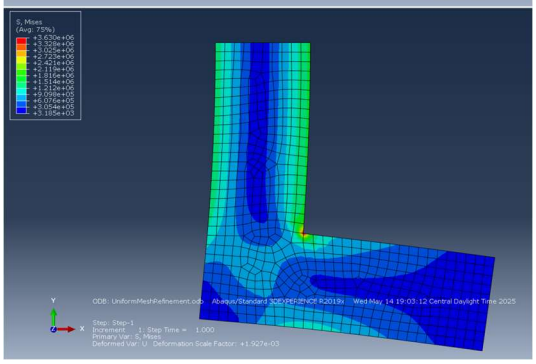
Now, we check for convergence of stresses by conducting a uniform h-refinement test. Just as before, the following images correspond to the entries in the data table. For the sake of space, I made the decision to remove the images of the structure before the deformation.

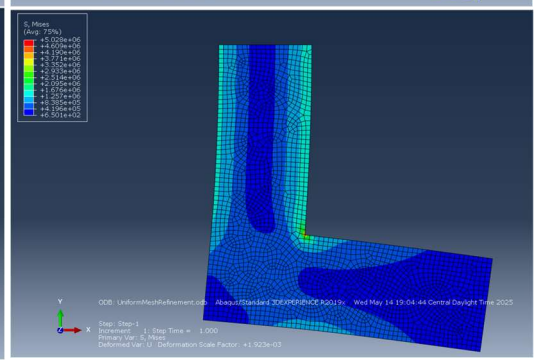


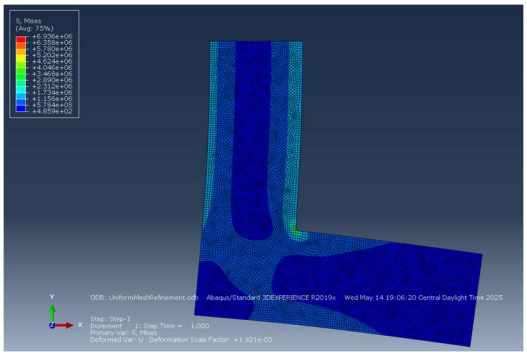



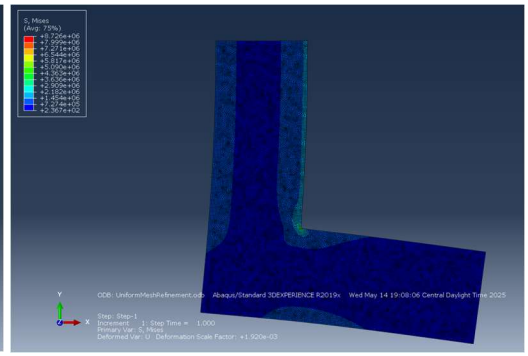



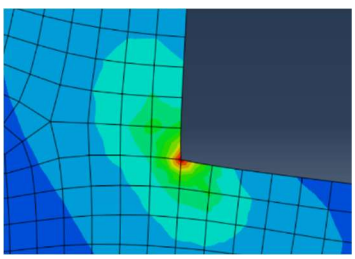



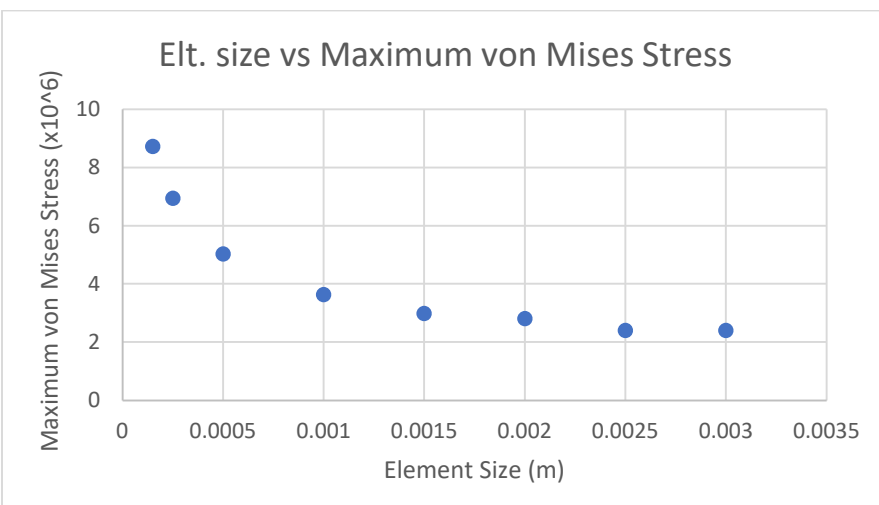




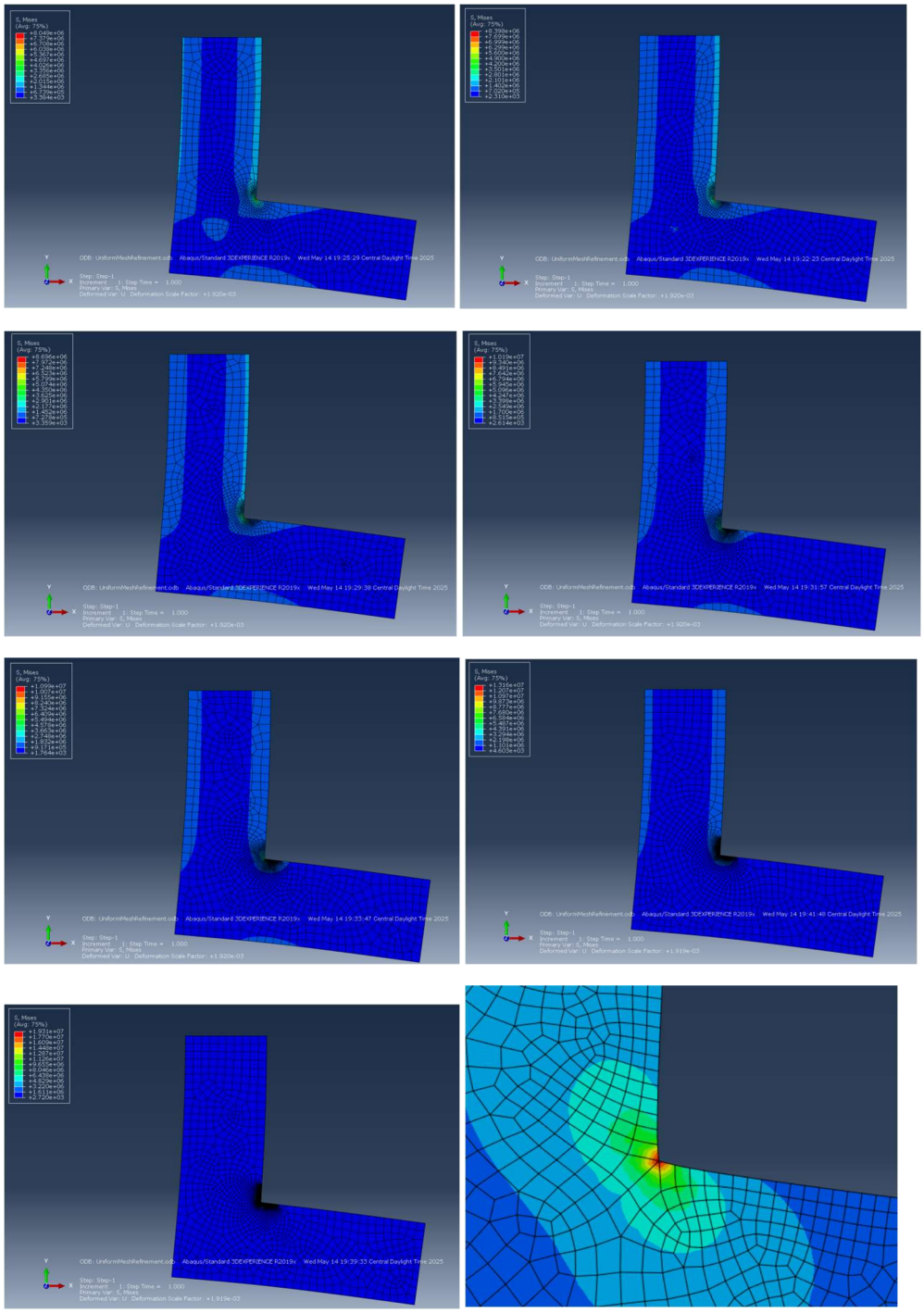
By conducting a uniform mesh refinement, we see that the maximum von Mises stress does not converge, meaning this geometry likely leads to a singularity at the inside left corner. See the image and data below. As the element size approaches 0, the maximum stress seems to approach infinity.



| Element size | Maximum von Mises stress (x10^6) |
|---|---|
| 0.003 | 2.395 |
| 0.0025 | 2.404 |
| 0.002 | 2.804 |
| 0.0015 | 2.98 |
| 0.001 | 3.63 |
| 0.0005 | 5.028 |
| 0.00025 | 6.936 |
| 0.00015 | 8.726 |

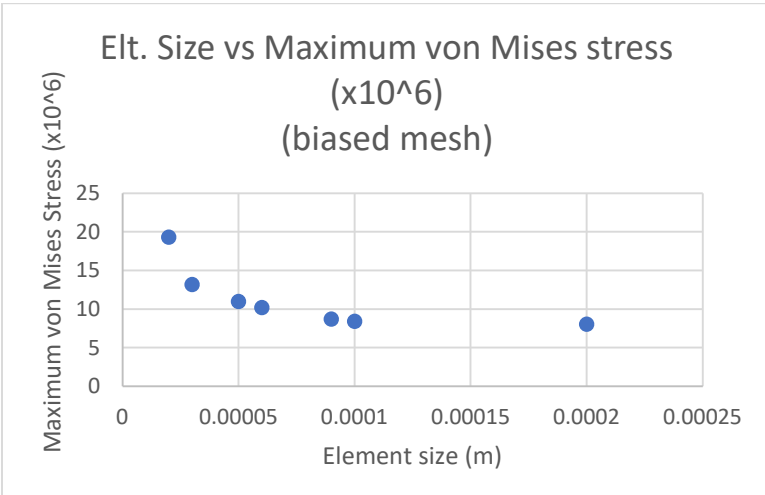



Now for a biased mesh refinement. We will keep the size of the rest of the elements constant (0.001). We partition the edges so we can control the size of elements close to the corner. Again, for the sake of space we only include the deformed shape.

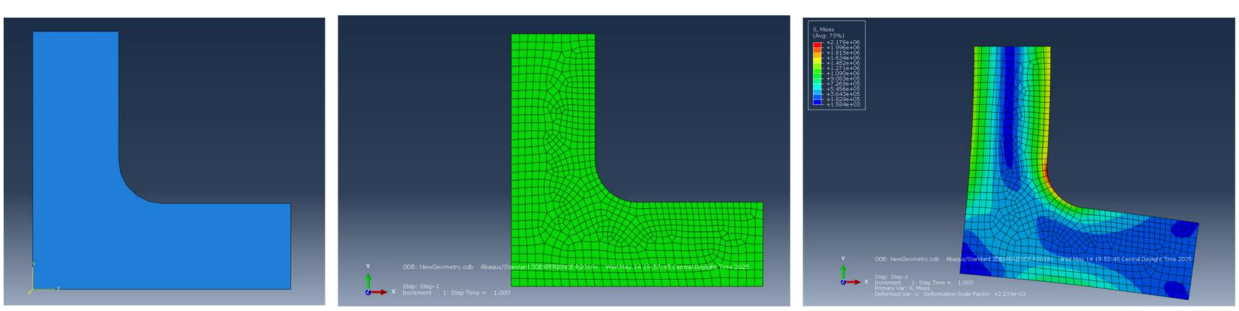


Once again, we see a singularity form at the reentrant corner, and no sign of convergence. See the data below. We conclude that the maximum von Mises stress **does not** converge with mesh refinement.

| Size of element near the corner | Maximum von Mises stress (x10^6) |
|---|---|
| 0.0002 | 8.049 |
| 0.0001 | 8.398 |
| 0.00009 | 8.696 |
| 0.00006 | 10.19 |
| 0.00005 | 10.99 |
| 0.00003 | 13.16 |
| 0.00002 | 19.31 |

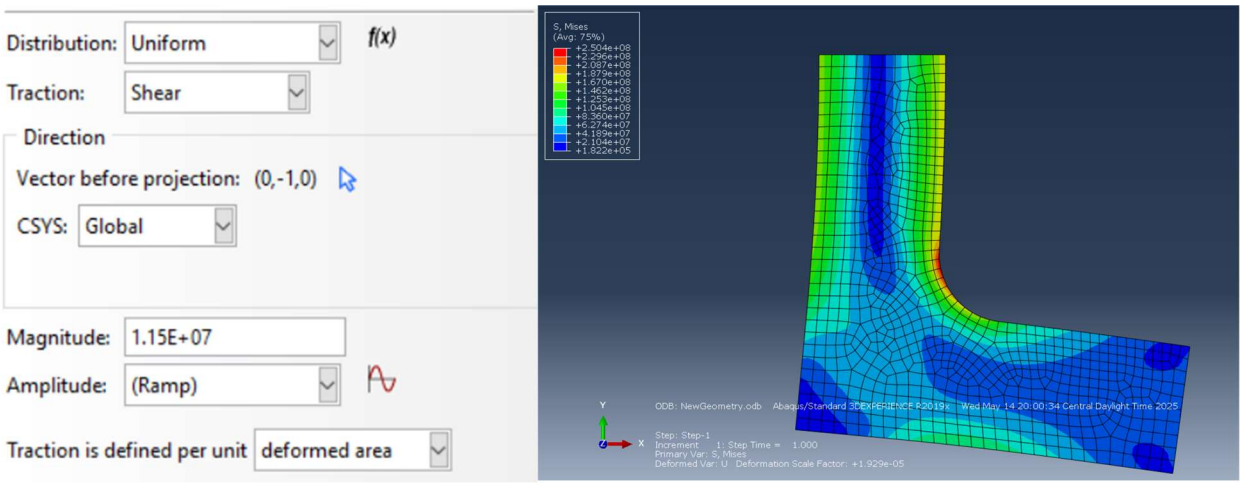



To handle this singularity, we propose a change in geometry. A possible shape refinement could be making the reentrant corner a smooth curve, instead of a 90-degree angle. This would likely disperse the stress along the edge, instead of isolating it right on the corner, thus avoiding the possibility of a singularity. See below:



The new geometry clearly handles the load much better, as the maximum stress here (2.178*10^6) is lower than *any* of the stresses from the many trials without the curve. Now to approximate the

maximum permissible load, I just ran through trial and error until I found a load that provided a stress larger than the yield stress (2.5 * 10^8 Pa) of this material. That load was: F = 1.15*10^5 N point load or (equivalently) F = 1.15*10^7 N distributed load.




This load yielded a maximum strain of 2.504*10^8 Pa, which surpasses the yield stress of this material.